

Sensitivity Analysis Audit Report

Project: Shadow AI Among Physicians in the German Healthcare System

1. Global Model Performance Evaluation Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only	N	Pseudo-R ²	AIC	BIC
0	0	0	0	320	0.1366	407.03	452.25
0	0	0	1	320	0.0922	377.37	422.59
0	0	1	0	320	0.1213	409.79	447.47
0	0	1	1	320	0.0891	374.56	412.25
0	1	0	0	320	0.1361	405.21	446.66
0	1	0	0	320	0.1361	405.21	446.66
0	1	0	1	320	0.0915	375.62	417.07
0	1	0	1	320	0.0915	375.62	417.07
0	1	1	0	320	0.1212	407.82	441.74
0	1	1	0	320	0.1212	407.82	441.74
0	1	1	1	320	0.0888	372.69	406.60
0	1	1	1	320	0.0888	372.69	406.60
1	0	0	0	320	0.1268	403.37	433.52
1	0	0	1	320	0.0720	377.22	407.36
1	0	1	0	320	0.1121	405.87	428.48
1	0	1	1	320	0.0685	374.59	397.20
1	1	0	0	320	0.1266	401.42	427.80
1	1	0	0	320	0.1266	401.42	427.80
1	1	0	1	320	0.0713	375.50	401.88
1	1	0	1	320	0.0713	375.50	401.88
1	1	1	0	320	0.1121	403.88	422.72
1	1	1	0	320	0.1121	403.88	422.72
1	1	1	1	320	0.0680	372.77	391.61
1	1	1	1	320	0.0680	372.77	391.61

2. Granular Model Interrogation Parameters

Model Specification 1

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.552	0.349	-1.579	0.114	0.576	0.291	1.142
KI-Bereitstellung	0.608	0.289	2.103	0.035	1.837	1.042	3.238
Guidelines	-0.948	0.507	-1.870	0.061	0.387	0.143	1.047
Schulung	0.380	0.362	1.051	0.293	1.462	0.720	2.971
Krankenhaus	0.526	0.297	1.770	0.077	1.691	0.945	3.026
MVZ	0.726	0.446	1.626	0.104	2.067	0.862	4.957
20-29 Jahre	2.201	0.814	2.702	0.007	9.033	1.831	44.576
30-39 Jahre	0.650	0.380	1.712	0.087	1.915	0.910	4.030
50-59 Jahre	-0.193	0.375	-0.515	0.606	0.824	0.396	1.718
60-69 Jahre	-1.085	0.440	-2.466	0.014	0.338	0.143	0.800
70 Jahre oder älter	-0.800	0.673	-1.188	0.235	0.449	0.120	1.681
männlich	0.077	0.260	0.297	0.767	1.080	0.649	1.798

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.880544
KI-Bereitstellung	1.145351
Guidelines	1.238844
Schulung	1.300186
Krankenhaus	1.545542
MVZ	1.135049
20-29 Jahre	1.453099
30-39 Jahre	2.238506
50-59 Jahre	1.967059
60-69 Jahre	1.771741
70 Jahre oder älter	1.301353
männlich	1.123089

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	60.575
LLR p-value	0.000
Pseudo R ² (McFadden)	0.137
AIC	407.027
BIC	452.247
Hosmer-Lemeshow p	0.396
Accuracy	0.647

Model Specification 2

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.208	0.379	-3.186	0.001	0.299	0.142	0.628
KI-Bereitstellung	0.380	0.291	1.305	0.192	1.463	0.826	2.589
Guidelines	-0.275	0.526	-0.523	0.601	0.759	0.271	2.128
Schulung	0.431	0.364	1.184	0.236	1.539	0.754	3.143
Krankenhaus	0.023	0.328	0.071	0.943	1.024	0.539	1.945
MVZ	0.278	0.488	0.569	0.569	1.320	0.507	3.436
20-29 Jahre	1.458	0.573	2.543	0.011	4.296	1.397	13.214
30-39 Jahre	0.142	0.398	0.357	0.721	1.153	0.528	2.516
50-59 Jahre	-0.338	0.402	-0.839	0.401	0.714	0.324	1.569
60-69 Jahre	-1.784	0.570	-3.132	0.002	0.168	0.055	0.513
70 Jahre oder älter	-0.793	0.734	-1.080	0.280	0.453	0.107	1.907
männlich	0.589	0.275	2.144	0.032	1.802	1.052	3.088

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.880544
KI-Bereitstellung	1.145351
Guidelines	1.238844
Schulung	1.300186
Krankenhaus	1.545542
MVZ	1.135049
20-29 Jahre	1.453099
30-39 Jahre	2.238506
50-59 Jahre	1.967059
60-69 Jahre	1.771741
70 Jahre oder älter	1.301353
männlich	1.123089

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	35.871
LLR p-value	0.000
Pseudo R ² (McFadden)	0.092
AIC	377.373
BIC	422.592
Hosmer-Lemeshow p	0.135
Accuracy	0.709

Model Specification 3

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.386	0.339	-1.139	0.255	0.680	0.350	1.320
Krankenhaus	0.503	0.293	1.717	0.086	1.654	0.931	2.939
MVZ	0.591	0.440	1.344	0.179	1.806	0.763	4.275
20-29 Jahre	2.119	0.809	2.618	0.009	8.326	1.704	40.686
30-39 Jahre	0.548	0.373	1.468	0.142	1.730	0.832	3.596
50-59 Jahre	-0.275	0.369	-0.746	0.455	0.759	0.368	1.565
60-69 Jahre	-1.174	0.434	-2.704	0.007	0.309	0.132	0.724
70 Jahre oder älter	-0.932	0.666	-1.398	0.162	0.394	0.107	1.454
männlich	0.079	0.257	0.308	0.758	1.082	0.654	1.792
Governance_Index	0.186	0.148	1.258	0.208	1.204	0.902	1.608

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.634360
Krankenhaus	1.541869
MVZ	1.125878
20-29 Jahre	1.444689
30-39 Jahre	2.221696
50-59 Jahre	1.953856
60-69 Jahre	1.749316
70 Jahre oder älter	1.289224
männlich	1.120927
Governance_Index	1.051131

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	53.813
LLR p-value	0.000
Pseudo R ² (McFadden)	0.121
AIC	409.788
BIC	447.472
Hosmer-Lemeshow p	0.484
Accuracy	0.656

Model Specification 4

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-0_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	0	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.132	0.370	-3.059	0.002	0.322	0.156	0.666
Krankenhaus	0.020	0.325	0.062	0.950	1.021	0.539	1.931
MVZ	0.203	0.486	0.417	0.677	1.225	0.472	3.175
20-29 Jahre	1.413	0.570	2.481	0.013	4.110	1.346	12.551
30-39 Jahre	0.099	0.394	0.252	0.801	1.104	0.510	2.391
50-59 Jahre	-0.377	0.399	-0.944	0.345	0.686	0.314	1.500
60-69 Jahre	-1.842	0.568	-3.243	0.001	0.159	0.052	0.483
70 Jahre oder älter	-0.871	0.729	-1.194	0.233	0.419	0.100	1.749
männlich	0.585	0.274	2.131	0.033	1.794	1.048	3.071
Governance_Index	0.259	0.156	1.662	0.097	1.296	0.955	1.760

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.634360
Krankenhaus	1.541869
MVZ	1.125878
20-29 Jahre	1.444689
30-39 Jahre	2.221696
50-59 Jahre	1.953856
60-69 Jahre	1.749316
70 Jahre oder älter	1.289224
männlich	1.120927
Governance_Index	1.051131

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	34.682
LLR p-value	0.000
Pseudo R ² (McFadden)	0.089
AIC	374.562
BIC	412.245
Hosmer-Lemeshow p	0.215
Accuracy	0.712

Model Specification 5

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.549	0.349	-1.572	0.116	0.578	0.291	1.145
KI-Bereitstellung	0.606	0.289	2.095	0.036	1.832	1.040	3.229
Guidelines	-0.924	0.502	-1.841	0.066	0.397	0.148	1.062
Schulung	0.383	0.361	1.060	0.289	1.467	0.722	2.978
20-29 Jahre	2.157	0.807	2.671	0.008	8.641	1.775	42.062
30-39 Jahre	0.609	0.367	1.658	0.097	1.839	0.895	3.778
50-59 Jahre	-0.202	0.374	-0.541	0.589	0.817	0.392	1.700
60-69 Jahre	-1.078	0.439	-2.456	0.014	0.340	0.144	0.804
70 Jahre oder älter	-0.816	0.672	-1.214	0.225	0.442	0.119	1.651
männlich	0.088	0.259	0.339	0.734	1.092	0.657	1.813
Krankenhaus und MVZ	0.576	0.272	2.118	0.034	1.778	1.044	3.030

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.878472
KI-Bereitstellung	1.145036
Guidelines	1.229960
Schulung	1.299202
20-29 Jahre	1.401998
30-39 Jahre	2.094331
50-59 Jahre	1.961064
60-69 Jahre	1.771610
70 Jahre oder älter	1.296737
männlich	1.111376
Krankenhaus und MVZ	1.305919

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	60.394
LLR p-value	0.000
Pseudo R ² (McFadden)	0.136
AIC	405.208
BIC	446.659
Hosmer-Lemeshow p	0.539
Accuracy	0.647

Model Specification 6

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.549	0.349	-1.572	0.116	0.578	0.291	1.145
KI-Bereitstellung	0.606	0.289	2.095	0.036	1.832	1.040	3.229
Guidelines	-0.924	0.502	-1.841	0.066	0.397	0.148	1.062
Schulung	0.383	0.361	1.060	0.289	1.467	0.722	2.978
20-29 Jahre	2.157	0.807	2.671	0.008	8.641	1.775	42.062
30-39 Jahre	0.609	0.367	1.658	0.097	1.839	0.895	3.778
50-59 Jahre	-0.202	0.374	-0.541	0.589	0.817	0.392	1.700
60-69 Jahre	-1.078	0.439	-2.456	0.014	0.340	0.144	0.804
70 Jahre oder älter	-0.816	0.672	-1.214	0.225	0.442	0.119	1.651
männlich	0.088	0.259	0.339	0.734	1.092	0.657	1.813
Krankenhaus und MVZ	0.576	0.272	2.118	0.034	1.778	1.044	3.030

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.878472
KI-Bereitstellung	1.145036
Guidelines	1.229960
Schulung	1.299202
20-29 Jahre	1.401998
30-39 Jahre	2.094331
50-59 Jahre	1.961064
60-69 Jahre	1.771610
70 Jahre oder älter	1.296737
männlich	1.111376
Krankenhaus und MVZ	1.305919

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	60.394
LLR p-value	0.000
Pseudo R ² (McFadden)	0.136
AIC	405.208
BIC	446.659
Hosmer-Lemeshow p	0.539
Accuracy	0.647

Model Specification 7

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.201	0.379	-3.170	0.002	0.301	0.143	0.632
KI-Bereitstellung	0.375	0.291	1.288	0.198	1.455	0.822	2.575
Guidelines	-0.237	0.518	-0.457	0.648	0.789	0.286	2.179
Schulung	0.435	0.364	1.193	0.233	1.544	0.756	3.154
20-29 Jahre	1.399	0.560	2.500	0.012	4.052	1.353	12.136
30-39 Jahre	0.088	0.383	0.231	0.817	1.092	0.516	2.313
50-59 Jahre	-0.351	0.401	-0.876	0.381	0.704	0.321	1.544
60-69 Jahre	-1.777	0.569	-3.123	0.002	0.169	0.055	0.516
70 Jahre oder älter	-0.815	0.732	-1.112	0.266	0.443	0.105	1.860
männlich	0.601	0.274	2.197	0.028	1.824	1.067	3.118
Krankenhaus und MVZ	0.085	0.302	0.281	0.778	1.089	0.602	1.969

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.878472
KI-Bereitstellung	1.145036
Guidelines	1.229960
Schulung	1.299202
20-29 Jahre	1.401998
30-39 Jahre	2.094331
50-59 Jahre	1.961064
60-69 Jahre	1.771610
70 Jahre oder älter	1.296737
männlich	1.111376
Krankenhaus und MVZ	1.305919

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	35.626
LLR p-value	0.000
Pseudo R ² (McFadden)	0.092
AIC	375.618
BIC	417.070
Hosmer-Lemeshow p	0.197
Accuracy	0.709

Model Specification 8

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.201	0.379	-3.170	0.002	0.301	0.143	0.632
KI-Bereitstellung	0.375	0.291	1.288	0.198	1.455	0.822	2.575
Guidelines	-0.237	0.518	-0.457	0.648	0.789	0.286	2.179
Schulung	0.435	0.364	1.193	0.233	1.544	0.756	3.154
20-29 Jahre	1.399	0.560	2.500	0.012	4.052	1.353	12.136
30-39 Jahre	0.088	0.383	0.231	0.817	1.092	0.516	2.313
50-59 Jahre	-0.351	0.401	-0.876	0.381	0.704	0.321	1.544
60-69 Jahre	-1.777	0.569	-3.123	0.002	0.169	0.055	0.516
70 Jahre oder älter	-0.815	0.732	-1.112	0.266	0.443	0.105	1.860
männlich	0.601	0.274	2.197	0.028	1.824	1.067	3.118
Krankenhaus und MVZ	0.085	0.302	0.281	0.778	1.089	0.602	1.969

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.878472
KI-Bereitstellung	1.145036
Guidelines	1.229960
Schulung	1.299202
20-29 Jahre	1.401998
30-39 Jahre	2.094331
50-59 Jahre	1.961064
60-69 Jahre	1.771610
70 Jahre oder älter	1.296737
männlich	1.111376
Krankenhaus und MVZ	1.305919

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	35.626
LLR p-value	0.000
Pseudo R ² (McFadden)	0.092
AIC	375.618
BIC	417.070
Hosmer-Lemeshow p	0.197
Accuracy	0.709

Model Specification 9

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.385	0.338	-1.138	0.255	0.680	0.350	1.321
20-29 Jahre	2.100	0.803	2.615	0.009	8.170	1.693	39.429
30-39 Jahre	0.531	0.362	1.468	0.142	1.700	0.837	3.454
50-59 Jahre	-0.279	0.368	-0.757	0.449	0.757	0.367	1.558
60-69 Jahre	-1.171	0.434	-2.701	0.007	0.310	0.133	0.725
70 Jahre oder älter	-0.938	0.665	-1.410	0.159	0.391	0.106	1.442
männlich	0.083	0.256	0.326	0.745	1.087	0.658	1.796
Krankenhaus und MVZ	0.525	0.268	1.958	0.050	1.691	0.999	2.862
Governance_Index	0.188	0.147	1.275	0.202	1.206	0.904	1.609

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.634296
20-29 Jahre	1.394211
30-39 Jahre	2.084304
50-59 Jahre	1.949137
60-69 Jahre	1.748828
70 Jahre oder älter	1.285334
männlich	1.109053
Krankenhaus und MVZ	1.300593
Governance_Index	1.046106

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	53.778
LLR p-value	0.000
Pseudo R ² (McFadden)	0.121
AIC	407.824
BIC	441.738
Hosmer-Lemeshow p	0.351
Accuracy	0.653

Model Specification 10

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.385	0.338	-1.138	0.255	0.680	0.350	1.321
20-29 Jahre	2.100	0.803	2.615	0.009	8.170	1.693	39.429
30-39 Jahre	0.531	0.362	1.468	0.142	1.700	0.837	3.454
50-59 Jahre	-0.279	0.368	-0.757	0.449	0.757	0.367	1.558
60-69 Jahre	-1.171	0.434	-2.701	0.007	0.310	0.133	0.725
70 Jahre oder älter	-0.938	0.665	-1.410	0.159	0.391	0.106	1.442
männlich	0.083	0.256	0.326	0.745	1.087	0.658	1.796
Krankenhaus und MVZ	0.525	0.268	1.958	0.050	1.691	0.999	2.862
Governance_Index	0.188	0.147	1.275	0.202	1.206	0.904	1.609

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.634296
20-29 Jahre	1.394211
30-39 Jahre	2.084304
50-59 Jahre	1.949137
60-69 Jahre	1.748828
70 Jahre oder älter	1.285334
männlich	1.109053
Krankenhaus und MVZ	1.300593
Governance_Index	1.046106

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	53.778
LLR p-value	0.000
Pseudo R ² (McFadden)	0.121
AIC	407.824
BIC	441.738
Hosmer-Lemeshow p	0.351
Accuracy	0.653

Model Specification 11

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.130	0.370	-3.054	0.002	0.323	0.156	0.667
20-29 Jahre	1.373	0.557	2.463	0.014	3.946	1.324	11.764
30-39 Jahre	0.062	0.380	0.164	0.870	1.064	0.505	2.241
50-59 Jahre	-0.386	0.398	-0.968	0.333	0.680	0.311	1.484
60-69 Jahre	-1.837	0.567	-3.238	0.001	0.159	0.052	0.484
70 Jahre oder älter	-0.885	0.728	-1.215	0.224	0.413	0.099	1.721
männlich	0.594	0.273	2.175	0.030	1.811	1.061	3.092
Krankenhaus und MVZ	0.064	0.300	0.215	0.830	1.067	0.592	1.922
Governance_Index	0.264	0.155	1.701	0.089	1.302	0.961	1.766

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.634296
20-29 Jahre	1.394211
30-39 Jahre	2.084304
50-59 Jahre	1.949137
60-69 Jahre	1.748828
70 Jahre oder älter	1.285334
männlich	1.109053
Krankenhaus und MVZ	1.300593
Governance_Index	1.046106

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	34.554
LLR p-value	0.000
Pseudo R ² (McFadden)	0.089
AIC	372.690
BIC	406.605
Hosmer-Lemeshow p	0.235
Accuracy	0.716

Model Specification 12

Source Identifier:
ordinal_age-0_hospital_and_mvz_group-1_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
0	1	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-1.130	0.370	-3.054	0.002	0.323	0.156	0.667
20-29 Jahre	1.373	0.557	2.463	0.014	3.946	1.324	11.764
30-39 Jahre	0.062	0.380	0.164	0.870	1.064	0.505	2.241
50-59 Jahre	-0.386	0.398	-0.968	0.333	0.680	0.311	1.484
60-69 Jahre	-1.837	0.567	-3.238	0.001	0.159	0.052	0.484
70 Jahre oder älter	-0.885	0.728	-1.215	0.224	0.413	0.099	1.721
männlich	0.594	0.273	2.175	0.030	1.811	1.061	3.092
Krankenhaus und MVZ	0.064	0.300	0.215	0.830	1.067	0.592	1.922
Governance_Index	0.264	0.155	1.701	0.089	1.302	0.961	1.766

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.634296
20-29 Jahre	1.394211
30-39 Jahre	2.084304
50-59 Jahre	1.949137
60-69 Jahre	1.748828
70 Jahre oder älter	1.285334
männlich	1.109053
Krankenhaus und MVZ	1.300593
Governance_Index	1.046106

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	34.554
LLR p-value	0.000
Pseudo R ² (McFadden)	0.089
AIC	372.690
BIC	406.605
Hosmer-Lemeshow p	0.235
Accuracy	0.716

Model Specification 13

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.715	0.358	1.995	0.046	2.044	1.013	4.125
KI-Bereitstellung	0.606	0.286	2.120	0.034	1.833	1.047	3.209
Guidelines	-0.898	0.500	-1.797	0.072	0.408	0.153	1.085
Schulung	0.320	0.358	0.892	0.372	1.377	0.682	2.779
Krankenhaus	0.541	0.286	1.891	0.059	1.718	0.980	3.010
MVZ	0.645	0.437	1.477	0.140	1.906	0.810	4.486
männlich	0.097	0.257	0.378	0.706	1.102	0.666	1.825
Alter	-0.533	0.111	-4.821	0.000	0.587	0.472	0.729

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.989178
KI-Bereitstellung	1.127611
Guidelines	1.229663
Schulung	1.272042
Krankenhaus	1.422571
MVZ	1.106831
männlich	1.107607
Alter	1.398117

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	56.230
LLR p-value	0.000
Pseudo R ² (McFadden)	0.127
AIC	403.372
BIC	433.519
Hosmer-Lemeshow p	0.239
Accuracy	0.656

Model Specification 14

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.225	0.387	-0.582	0.561	0.798	0.374	1.705
KI-Bereitstellung	0.427	0.286	1.492	0.136	1.532	0.875	2.683
Guidelines	-0.282	0.514	-0.549	0.583	0.754	0.275	2.066
Schulung	0.380	0.356	1.065	0.287	1.462	0.727	2.940
Krankenhaus	-0.067	0.314	-0.212	0.832	0.935	0.506	1.731
MVZ	0.194	0.470	0.414	0.679	1.215	0.484	3.050
männlich	0.551	0.270	2.037	0.042	1.734	1.021	2.945
Alter	-0.510	0.122	-4.181	0.000	0.600	0.473	0.763

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.989178
KI-Bereitstellung	1.127611
Guidelines	1.229663
Schulung	1.272042
Krankenhaus	1.422571
MVZ	1.106831
männlich	1.107607
Alter	1.398117

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	28.028
LLR p-value	0.000
Pseudo R ² (McFadden)	0.072
AIC	377.216
BIC	407.362
Hosmer-Lemeshow p	0.012
Accuracy	0.691

Model Specification 15

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.821	0.354	2.320	0.020	2.272	1.136	4.545
Krankenhaus	0.505	0.282	1.792	0.073	1.658	0.954	2.881
MVZ	0.531	0.431	1.233	0.218	1.700	0.731	3.953
männlich	0.086	0.254	0.337	0.736	1.090	0.662	1.794
Alter	-0.537	0.110	-4.899	0.000	0.584	0.471	0.724
Governance_Index	0.178	0.145	1.226	0.220	1.194	0.899	1.586

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.829719
Krankenhaus	1.415650
MVZ	1.098941
männlich	1.104588
Alter	1.391845
Governance_Index	1.022993

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	49.728
LLR p-value	0.000
Pseudo R ² (McFadden)	0.112
AIC	405.874
BIC	428.484
Hosmer-Lemeshow p	0.353
Accuracy	0.656

Model Specification 16

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-0_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	0	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.170	0.383	-0.446	0.656	0.843	0.398	1.785
Krankenhaus	-0.076	0.312	-0.243	0.808	0.927	0.503	1.709
MVZ	0.131	0.469	0.279	0.780	1.140	0.455	2.857
männlich	0.539	0.270	2.001	0.045	1.715	1.011	2.909
Alter	-0.514	0.122	-4.226	0.000	0.598	0.471	0.759
Governance_Index	0.260	0.152	1.714	0.087	1.297	0.963	1.747

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.829719
Krankenhaus	1.415650
MVZ	1.098941
männlich	1.104588
Alter	1.391845
Governance_Index	1.022993

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	26.652
LLR p-value	0.000
Pseudo R ² (McFadden)	0.068
AIC	374.592
BIC	397.202
Hosmer-Lemeshow p	0.042
Accuracy	0.709

Model Specification 17

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.694	0.346	2.004	0.045	2.002	1.015	3.947
KI-Bereitstellung	0.606	0.286	2.121	0.034	1.833	1.047	3.210
Guidelines	-0.887	0.496	-1.787	0.074	0.412	0.156	1.090
Schulung	0.323	0.358	0.901	0.367	1.381	0.685	2.785
männlich	0.102	0.256	0.398	0.691	1.107	0.670	1.830
Alter	-0.527	0.107	-4.917	0.000	0.590	0.478	0.728
Krankenhaus und MVZ	0.566	0.264	2.146	0.032	1.761	1.050	2.954

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.434733
KI-Bereitstellung	1.127564
Guidelines	1.222575
Schulung	1.269247
männlich	1.100250
Alter	1.322171
Krankenhaus und MVZ	1.226784

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	56.178
LLR p-value	0.000
Pseudo R ² (McFadden)	0.127
AIC	401.424
BIC	427.802
Hosmer-Lemeshow p	0.198
Accuracy	0.656

Model Specification 18

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.694	0.346	2.004	0.045	2.002	1.015	3.947
KI-Bereitstellung	0.606	0.286	2.121	0.034	1.833	1.047	3.210
Guidelines	-0.887	0.496	-1.787	0.074	0.412	0.156	1.090
Schulung	0.323	0.358	0.901	0.367	1.381	0.685	2.785
männlich	0.102	0.256	0.398	0.691	1.107	0.670	1.830
Alter	-0.527	0.107	-4.917	0.000	0.590	0.478	0.728
Krankenhaus und MVZ	0.566	0.264	2.146	0.032	1.761	1.050	2.954

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.434733
KI-Bereitstellung	1.127564
Guidelines	1.222575
Schulung	1.269247
männlich	1.100250
Alter	1.322171
Krankenhaus und MVZ	1.226784

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	56.178
LLR p-value	0.000
Pseudo R ² (McFadden)	0.127
AIC	401.424
BIC	427.802
Hosmer-Lemeshow p	0.198
Accuracy	0.656

Model Specification 19

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.279	0.374	-0.747	0.455	0.756	0.364	1.573
KI-Bereitstellung	0.426	0.286	1.491	0.136	1.532	0.875	2.682
Guidelines	-0.250	0.509	-0.491	0.623	0.779	0.287	2.111
Schulung	0.386	0.356	1.083	0.279	1.471	0.732	2.958
männlich	0.562	0.269	2.087	0.037	1.754	1.035	2.975
Alter	-0.494	0.118	-4.197	0.000	0.610	0.485	0.769
Krankenhaus und MVZ	-0.005	0.291	-0.017	0.987	0.995	0.563	1.760

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.434733
KI-Bereitstellung	1.127564
Guidelines	1.222575
Schulung	1.269247
männlich	1.100250
Alter	1.322171
Krankenhaus und MVZ	1.226784

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	27.746
LLR p-value	0.000
Pseudo R ² (McFadden)	0.071
AIC	375.497
BIC	401.875
Hosmer-Lemeshow p	0.188
Accuracy	0.694

Model Specification 20

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-0_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	0	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.279	0.374	-0.747	0.455	0.756	0.364	1.573
KI-Bereitstellung	0.426	0.286	1.491	0.136	1.532	0.875	2.682
Guidelines	-0.250	0.509	-0.491	0.623	0.779	0.287	2.111
Schulung	0.386	0.356	1.083	0.279	1.471	0.732	2.958
männlich	0.562	0.269	2.087	0.037	1.754	1.035	2.975
Alter	-0.494	0.118	-4.197	0.000	0.610	0.485	0.769
Krankenhaus und MVZ	-0.005	0.291	-0.017	0.987	0.995	0.563	1.760

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.434733
KI-Bereitstellung	1.127564
Guidelines	1.222575
Schulung	1.269247
männlich	1.100250
Alter	1.322171
Krankenhaus und MVZ	1.226784

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	27.746
LLR p-value	0.000
Pseudo R ² (McFadden)	0.071
AIC	375.497
BIC	401.875
Hosmer-Lemeshow p	0.188
Accuracy	0.694

Model Specification 21

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.816	0.342	2.386	0.017	2.261	1.157	4.417
männlich	0.087	0.254	0.343	0.732	1.091	0.664	1.793
Alter	-0.536	0.107	-5.029	0.000	0.585	0.475	0.721
Krankenhaus und MVZ	0.511	0.260	1.968	0.049	1.668	1.002	2.776
Governance_Index	0.178	0.144	1.237	0.216	1.195	0.901	1.585

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.241200
männlich	1.097114
Alter	1.313809
Krankenhaus und MVZ	1.218331
Governance_Index	1.013063

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	49.725
LLR p-value	0.000
Pseudo R² (McFadden)	0.112
AIC	403.877
BIC	422.719
Hosmer-Lemeshow p	0.342
Accuracy	0.656

Model Specification 22

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-0_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	0

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	0.816	0.342	2.386	0.017	2.261	1.157	4.417
männlich	0.087	0.254	0.343	0.732	1.091	0.664	1.793
Alter	-0.536	0.107	-5.029	0.000	0.585	0.475	0.721
Krankenhaus und MVZ	0.511	0.260	1.968	0.049	1.668	1.002	2.776
Governance_Index	0.178	0.144	1.237	0.216	1.195	0.901	1.585

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.241200
männlich	1.097114
Alter	1.313809
Krankenhaus und MVZ	1.218331
Governance_Index	1.013063

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	49.725
LLR p-value	0.000
Pseudo R² (McFadden)	0.112
AIC	403.877
BIC	422.719
Hosmer-Lemeshow p	0.342
Accuracy	0.656

Model Specification 23

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-1.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.214	0.368	-0.583	0.560	0.807	0.392	1.660
männlich	0.550	0.269	2.047	0.041	1.733	1.024	2.933
Alter	-0.501	0.118	-4.266	0.000	0.606	0.481	0.763
Krankenhaus und MVZ	-0.027	0.289	-0.093	0.926	0.973	0.552	1.716
Governance_Index	0.267	0.151	1.770	0.077	1.306	0.972	1.755

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.241200
männlich	1.097114
Alter	1.313809
Krankenhaus und MVZ	1.218331
Governance_Index	1.013063

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	26.474
LLR p-value	0.000
Pseudo R² (McFadden)	0.068
AIC	372.769
BIC	391.611
Hosmer-Lemeshow p	0.206
Accuracy	0.709

Model Specification 24

Source Identifier:
ordinal_age-1_hospital_and_mvz_group-1_governance_index-1_only_ai_users-1_no_others_in_type_of_provider-0.xlsx

Hyperparameter Configuration Matrix

Age (Ord)	Hosp/MVZ Bundle	Gov Index	AI Users Only
1	1	1	1

Observations Sample Footprint: N = 320 valid cases included.

Estimated Coefficients & Target Odds Ratios (OR)

Unnamed: 0	Coef.	Std.Err.	z	P> z	Odds Ratio (OR)	OR 2.5%	OR 97.5%
const	-0.214	0.368	-0.583	0.560	0.807	0.392	1.660
männlich	0.550	0.269	2.047	0.041	1.733	1.024	2.933
Alter	-0.501	0.118	-4.266	0.000	0.606	0.481	0.763
Krankenhaus und MVZ	-0.027	0.289	-0.093	0.926	0.973	0.552	1.716
Governance_Index	0.267	0.151	1.770	0.077	1.306	0.972	1.755

Multicollinearity Diagnostics (VIF)

Variable	VIF
const	8.241200
männlich	1.097114
Alter	1.313809
Krankenhaus und MVZ	1.218331
Governance_Index	1.013063

Goodness of Fit Diagnostics

Metric	Value
LLR Statistic	26.474
LLR p-value	0.000
Pseudo R² (McFadden)	0.068
AIC	372.769
BIC	391.611
Hosmer-Lemeshow p	0.206
Accuracy	0.709